23-200-0105A
Basic Mechanical Engineering

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Division of Mechanical Engineering

Syllabus

Module I

Introduction: Role of Mechanical Engineering in Industries and Society- Emerging Trends and Technologies in different sectors such as Energy, Manufacturing, Automotive, Aerospace, and Marine sectors.

Introduction to thermodynamic laws, power generating devices: Boilers, Turbines (Steam & Gas), IC engines: Components and Working Principles of two stroke petrol engine and 4-Strokes Petrol and Diesel Engines, Application of IC Engines. (Elementary ideas only no numerical problems).

Introduction to power consuming devices: Refrigerator, types and properties of refrigerants, working of domestic refrigerators, Air-conditioning systems, Windows and Split systems (only elementary ideas, no numerical problems)

Module II

Introduction to power transmission systems: Belts, chain, and Gear drives, types and application, (numerical problems related to simple power calculations only).

Energy: Introduction and applications of Energy sources like Fossil fuels, nuclear fuels, Hydel, Solar, wind, and bio-fuels, Environmental issues like Global warming and Ozone depletion.

Modern fuel injection systems in CI and SI engines: CRDi, MPFI systems, cooling and lubricating systems in two stroke and four stroke engines. (Only elementary ideas with block diagrams).

Insight into Future Mobility: Electric and Hybrid Vehicles, Components of Electric and Hybrid Vehicles. Advantages and disadvantages of EVs and Hybrid vehicles.

Module III

Introduction to engineering materials: composite and smart materials.

Joining Processes: Soldering, Brazing and Welding, Definitions, classification of welding process, Arc welding, Gas welding and types of flames.

Machine Tool Operations: Working Principle of lathe, Lathe operations: Turning, facing, knurling. Working principles of Drilling Machine, drilling operations: drilling, boring, reaming. Working of Milling Machine, Milling operations: plane milling and slot milling.

(No sketches of machine tools, sketches to be used only for explaining the operations).

Introduction to Advanced Manufacturing Systems: Introduction, components of CNC, advantages and applications of CNC, 3D printing.

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Syllabus…

Module IV

Introduction to Mechatronics and Robotics: open-loop and closed-loop mechatronic systems. Classification based on robotics configuration: polar cylindrical, Cartesian coordinate and spherical. Application, Advantages and disadvantages.

Automation in industry: Definition, types – Fixed, programmable and flexible automation, basic elements with block diagrams, advantages.

Introduction to IOT: Definition and Characteristics, Physical design, protocols, Logical design of IoT, Functional blocks, and communication models.

References:

1. Jonathan Wickert and Kemper Lewis. An Introduction to Mechanical Engineering, Third Edition, Cengage Learning (2012).
2. Hazra Choudhry and Nirzar Roy. Elements of Workshop Technology (Vol. 1 and 2), Media Promoters and Publishers Pvt. Ltd. (2010).
3. V. Ganesan. Internal Combustion Engines, 4th edition, Tata McGraw Hill Education (2017).
4. Appu Kuttan K K. Robotics Volume 1, I K. International Publishing House Pvt Ltd (2013).
5. SRN Reddy, Rachit Thukral and Manasi Mishra. Introduction to Internet of Things: A Practical Approach, ETI Labs (2021).

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Joining Process

Joining Process

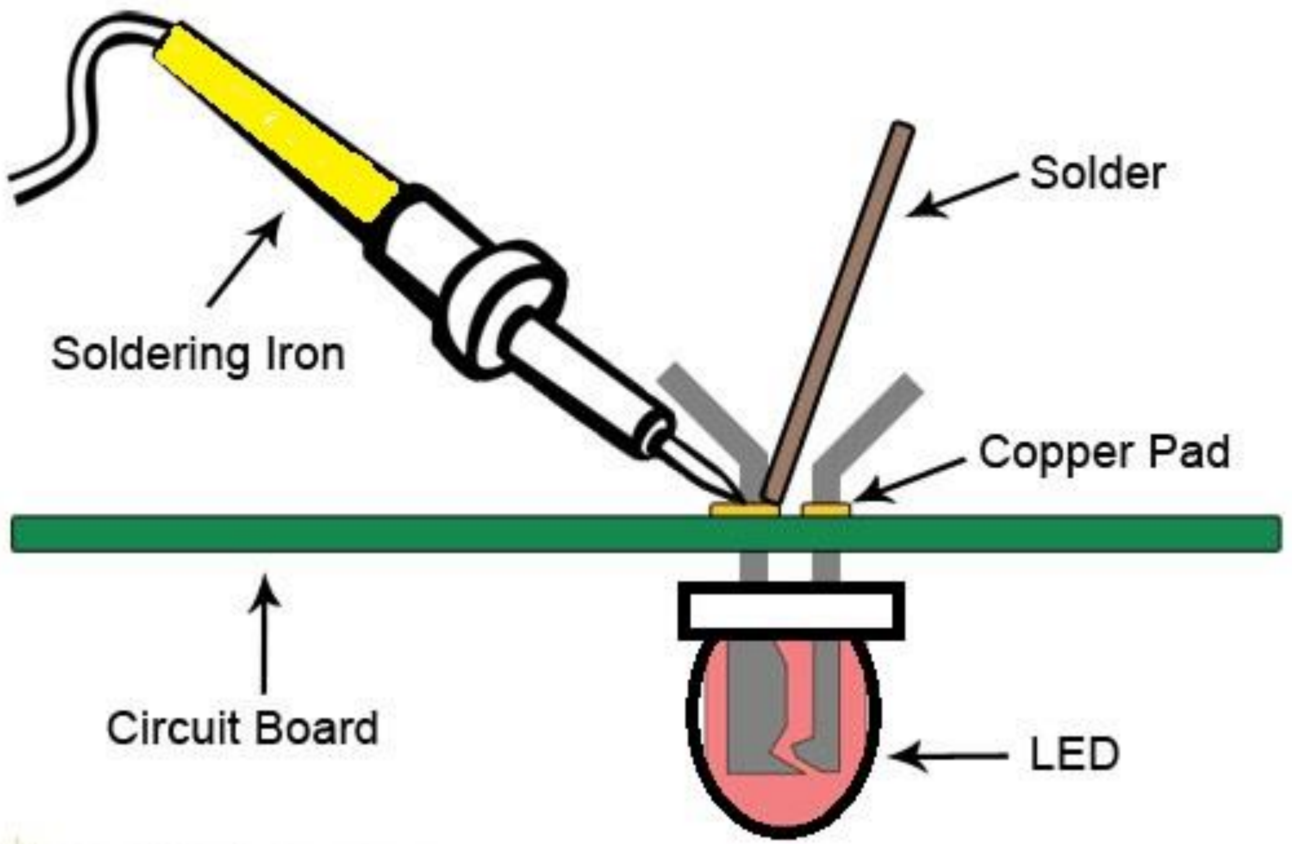
- Soldering
- Brazing
- Welding

Soldering Process

Soldering Process

- Soldering is a process in which two or more items are joined together by melting and putting a filler metal (Solder: Lead-tin alloy) into the joint.
- The melting temperature of filler metal is lower than 450°C and also lower than the melting point of the components to be joined.
- Unlike welding, soldering does not involve melting the work pieces.
- Solder is melted by using a soldering iron, which is heated by electrical resistance.
- The filler metal flows into the gap between close-fitting parts by capillary action.

Soldering Process



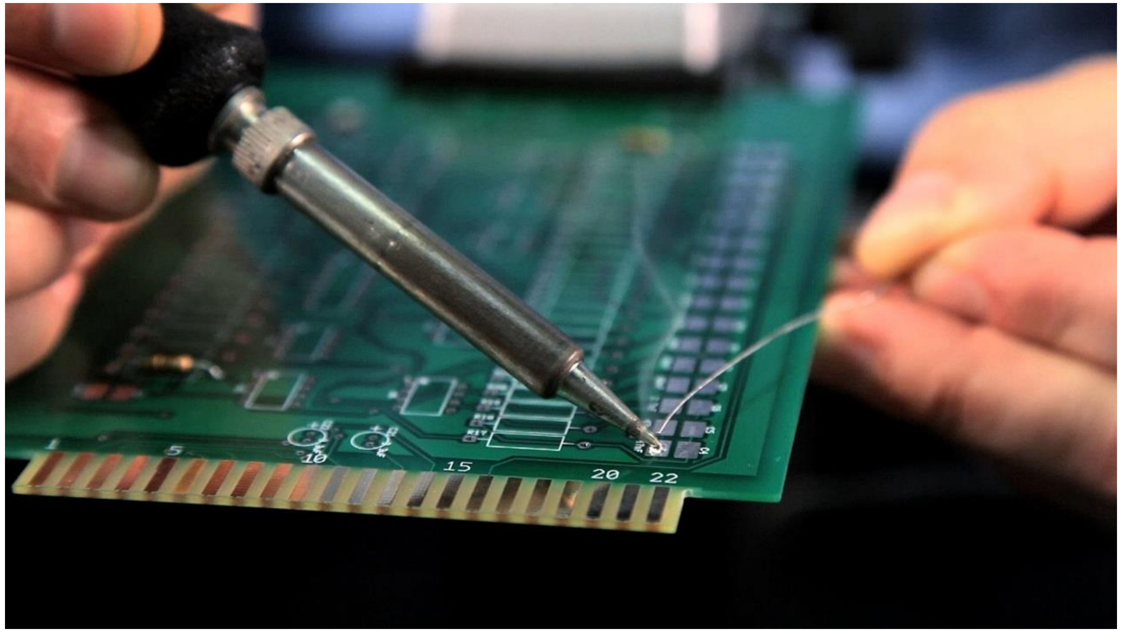
Applications

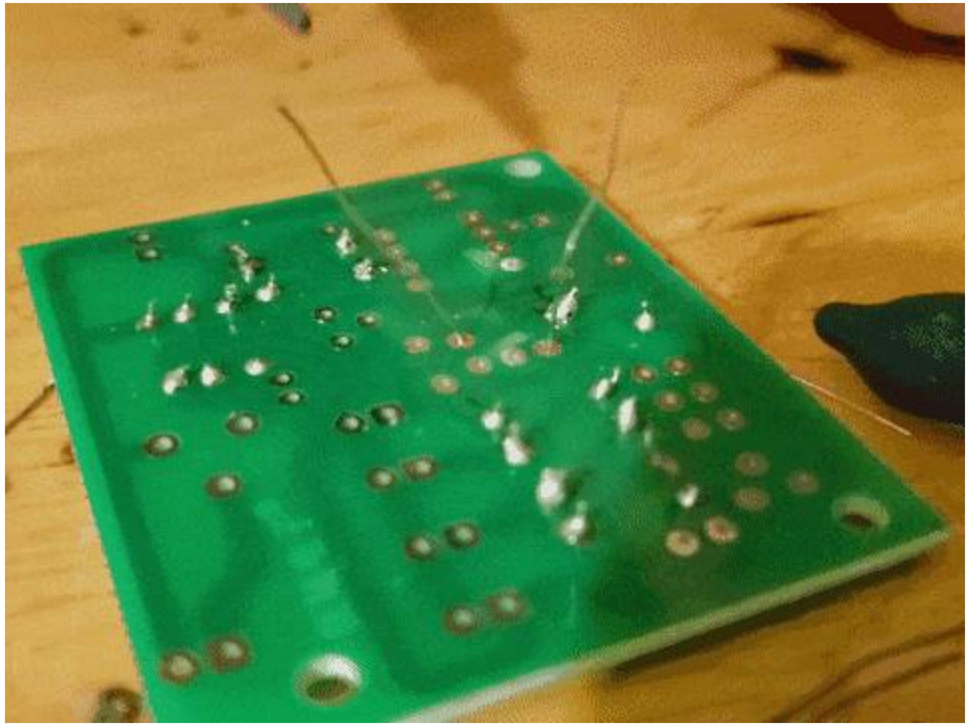
- It is mainly used in electronics industry -Electronic boards and similar small parts which are not subjected to load and temperature.
- Copper, silver and gold are easy to solder
- Electronics manufacture, implantable devices, medical devices, connectors etc.

Heating methods

- Electric soldering irons or guns
- Dip soldering (e.g. electronics work, automobile radiators)
- Induction heating (Used for a large number of identical parts)
- Heating by infrared sources (For low melting point solders)
- Furnace heating (Seldom used)

Example: Soldering

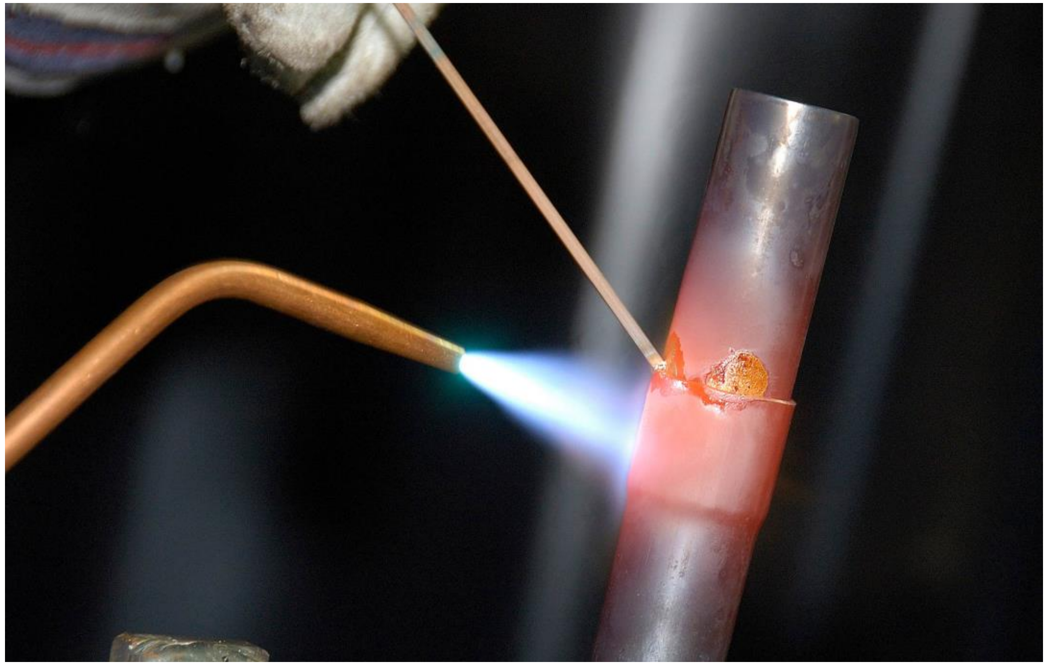


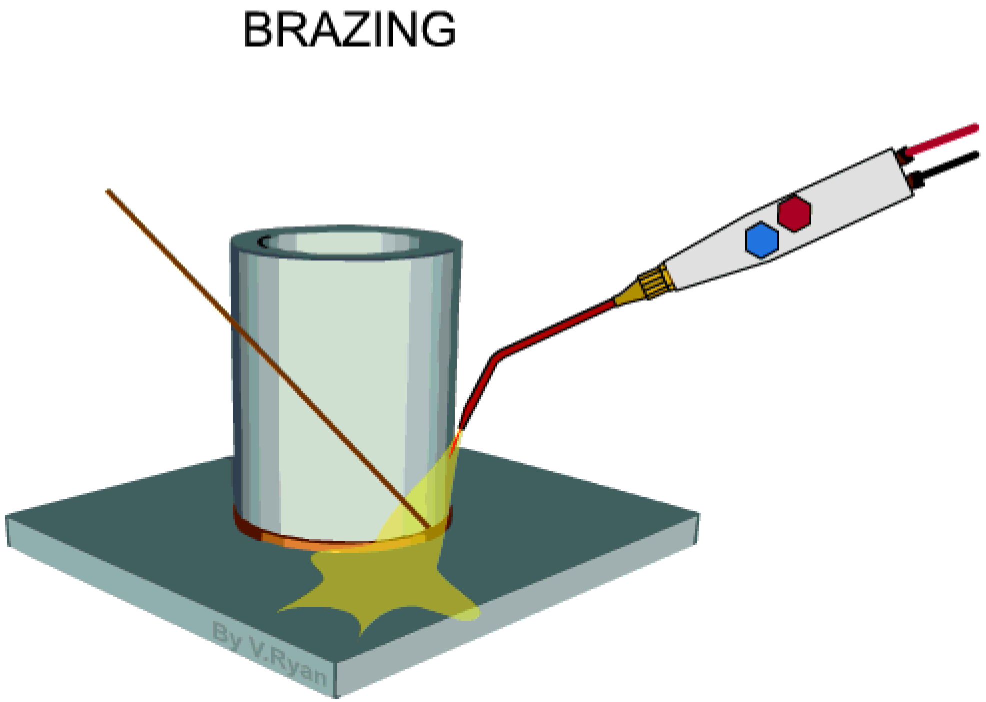


Brazing Process

- Brazing is a metal joining process in which two or more metal items are joined together by melting and flowing a filler metal (*copper-zinc alloy*) into the joint.
- The filler metal flows into the gap between close-fitting parts by capillary action. Cool it and solidify it.
- Melting temperature of filler metal is more than 450°C but lower than the melting temperature of the components to be joined.
- Brazing differs from welding in that it does not involve melting the work pieces.
- Brazing gives a much stronger joint compared to soldering

Example: Brazing





Materials

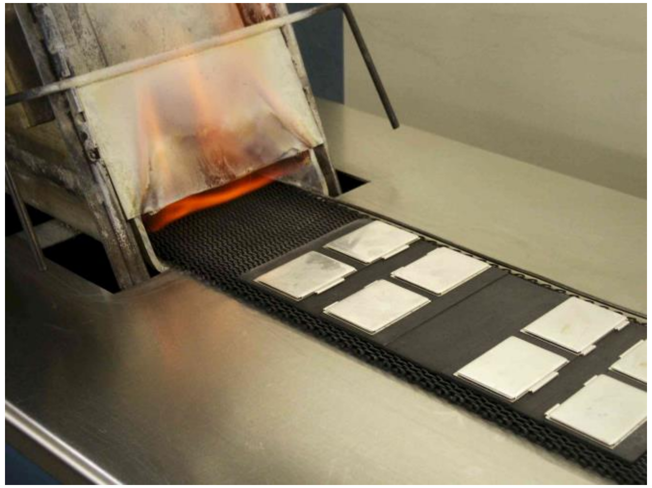
- Braze materials are:
  - Copper (for brazing of steel, HSS, and tungsten carbide),
  - Copper alloys (e.g. brazing brass, manganese bronze),
  - Silver (for brazing titanium),
  - Silver alloys,
  - Aluminium-silicon alloys

Heating Methods

- Torch brazing
- Furnace brazing
- Dip brazing
- Induction brazing
- Resistance brazing

Torch Brazing

Dip Brazing



Furnace Brazing

Advantages of Brazing over welding

- Virtually all metals can be joined.
- Less heating is required than for welding; the process can be performed more quickly and more economically.
- Since lower temperatures are used, fewer difficulties due to distortion are encountered, and thinner and more complex assemblies can be joined successfully.
- Many brazing operations are adaptable to automation

Sl No	Soldering	Brazing
1	Filler metal has a melting point below 450 degree Celsius	Filler alloy has melting point above 450 degree Celsius
2	Produces weaker joints than brazing	Produces stronger joints
3	Soldering joints do not resist corrosion	Brazed joints resist corrosion
4	Suitable for thin similar or dissimilar sheet metals	Suitable even for thicker dissimilar metals
5	Cost is less	Cost is more
6	A soldering iron or small blow torch is necessary	A furnace or heavy blow torch is necessary

Brazing and soldering

- Brazing and soldering are somewhat similar processes with some common properties as given below.
- The compositions of the brazing and soldering alloys, which are nonferrous alloys, are significantly different from the base metals (materials that are going to be joined).
- The melting point of these alloys are lower than that of the base metal.
- Therefore, during the process the base metals are not melted.
- Heating a brazed or soldered joint above the melting point of the used alloy may destroy the integrity of the joint

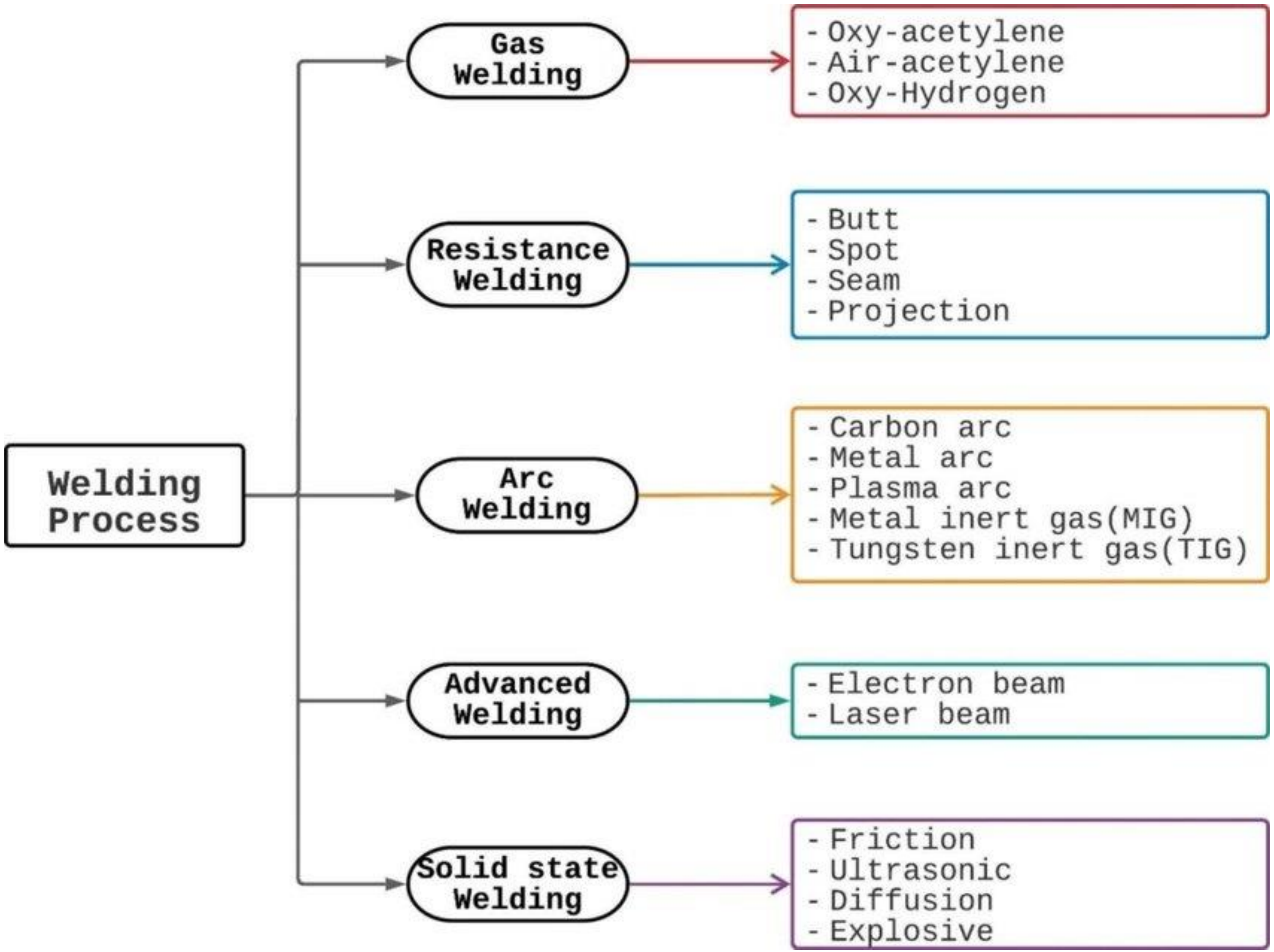
Welding

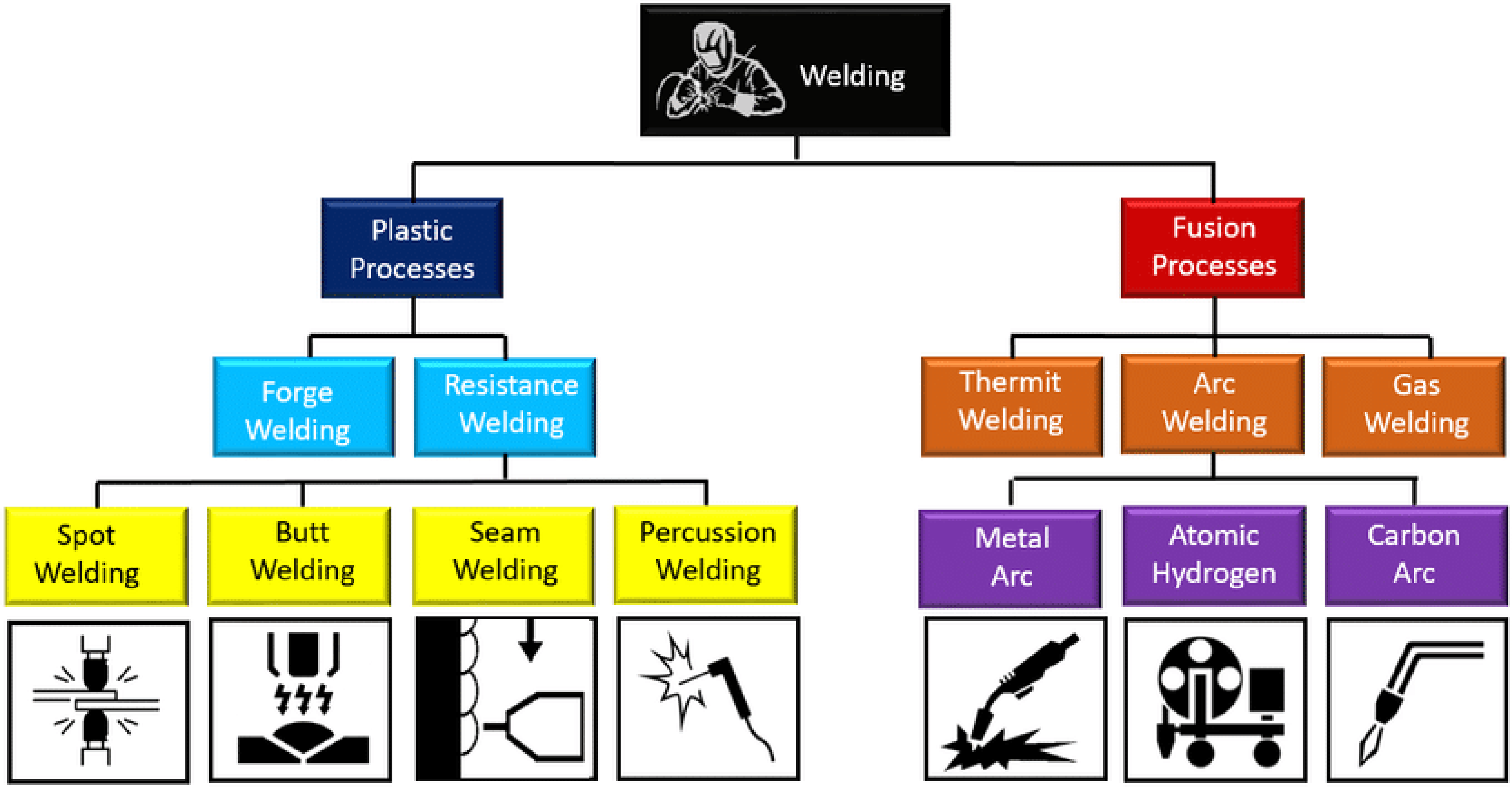
Welding Process

- Welding is the process of joining similar or dissimilar metals by the application of heat, with or without the application of pressure and addition of filler material
- Most essential requirement is heat.
- Pressure may or may not be used.
- Filler material may or may not be used.
- Flux is used to resist oxidation and to remove impurities.

Classification

- Plastic Welding -In this type of welding the metals to be joined are to be heated to the plastic state and then forced together by external pressure without the addition of filler material.
  - Eg. Forge Welding, resistance welding
- Fusion Welding:- In this type of welding no pressure is involved but a very high temperature is produced in or near the joint.
  - The metal at the joint is heated to the molten state and a joint is formed as a result of solidification.
  - The heat may be generated by electric arc, combustion of gases or chemical action. A filler material may be used during the welding process


Welding Process
Gas Welding
- Oxy-acetylene
- Air-acetylene
- Oxy-Hydrogen
Resistance Welding
- Butt
- Spot
- Seam
- Projection
Arc Welding
- Carbon arc
- Metal arc
- Plasma arc
- Metal inert gas(MIG)
- Tungsten inert gas(TIG)
Advanced Welding
- Electron beam
- Laser beam
Solid state Welding
- Friction
- Ultrasonic
- Diffusion
- Explosive

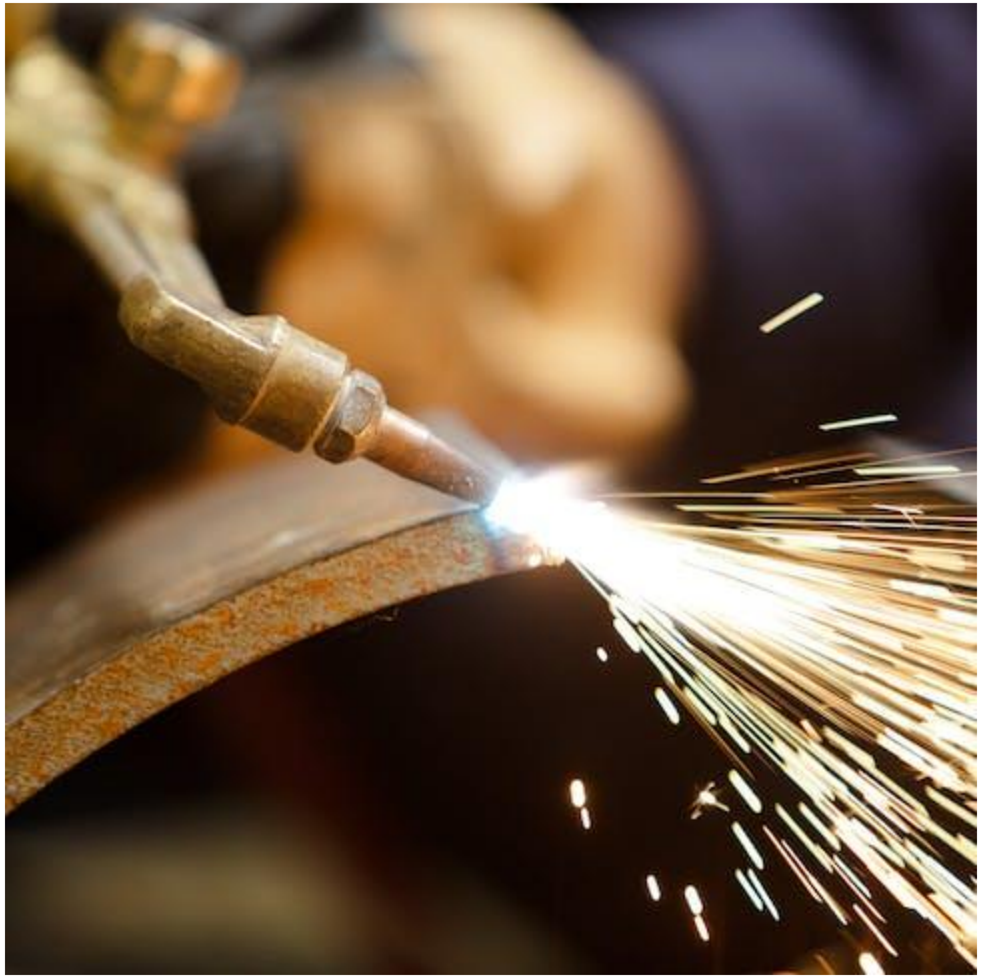

Welding
Plastic Processes
Fusion Processes
Forge Welding
Resistance Welding
Thermit Welding
Arc Welding
Gas Welding
Spot Welding
Butt Welding
Seam Welding
Percussion Welding
Metal Arc
Atomic Hydrogen
Carbon Arc

Applications

- Aircraft construction
- Automobile Construction (chassis, body, structure, doors)
- Bridges
- Railways (tracks, coaches/wagons).
- Buildings
- Pressure vessels and tanks
- Piping and Pipelines
- Ships
- Machine tool
- Household
- Heavy machineries

Gas Welding

- Gas Welding is a Fusion-Welding process.
- It joins metals using the heat of combustion of some fuel gas (acetylene, hydrogen, natural gas) in air or oxygen.
- Temperature produced ranges from 3000°C to 3400 °C
- The intense heat produced melts and fuses together the edges of parts to be welded, generally with the addition of filler material.
- It is used for welding ferrous and non ferrous metals particularly for thin sections up to 6 mm thick.



Gas Welding: Oxyacetylene welding

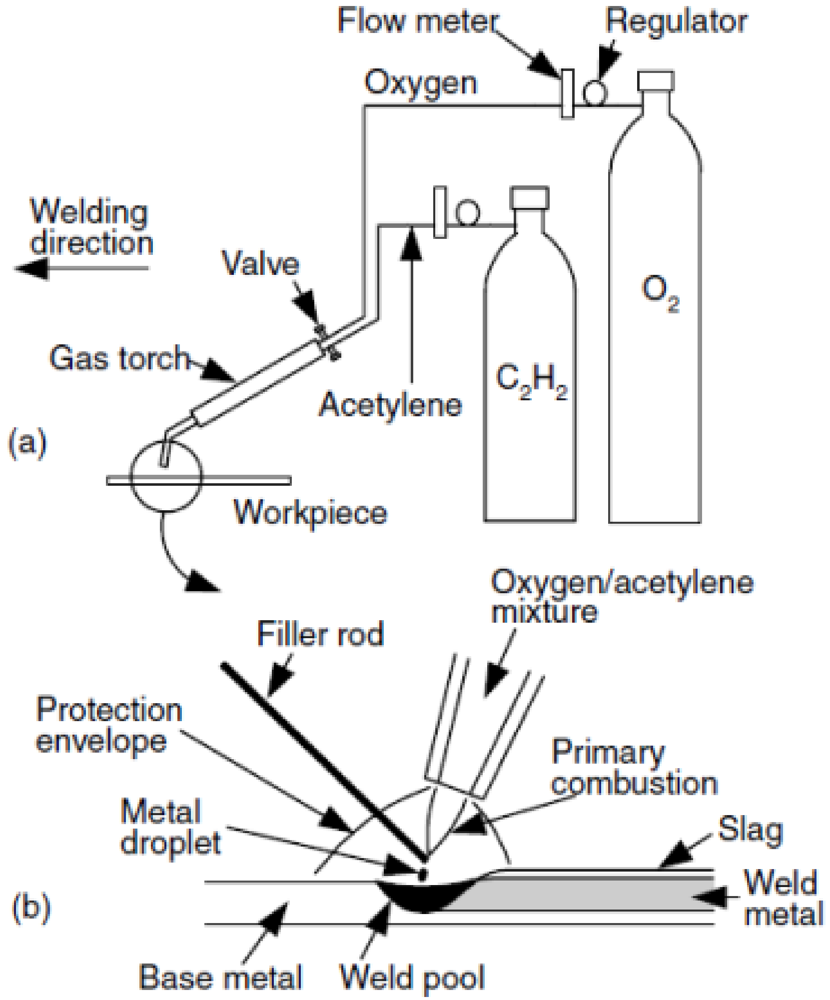
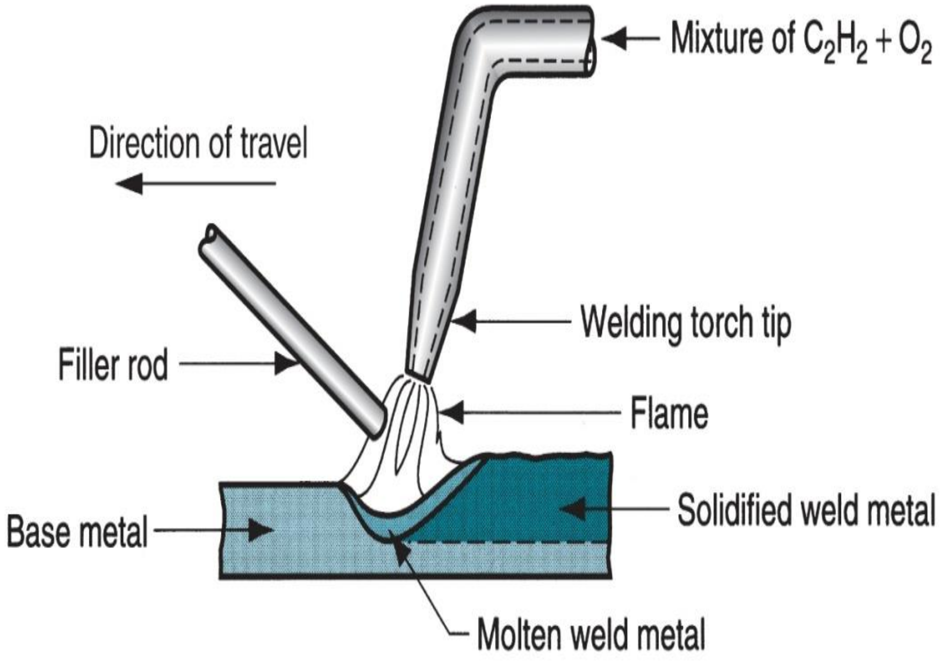


Figure 3.1 Oxyacetylene welding: (*a*) overall process; (*b*) welding area enlarged.

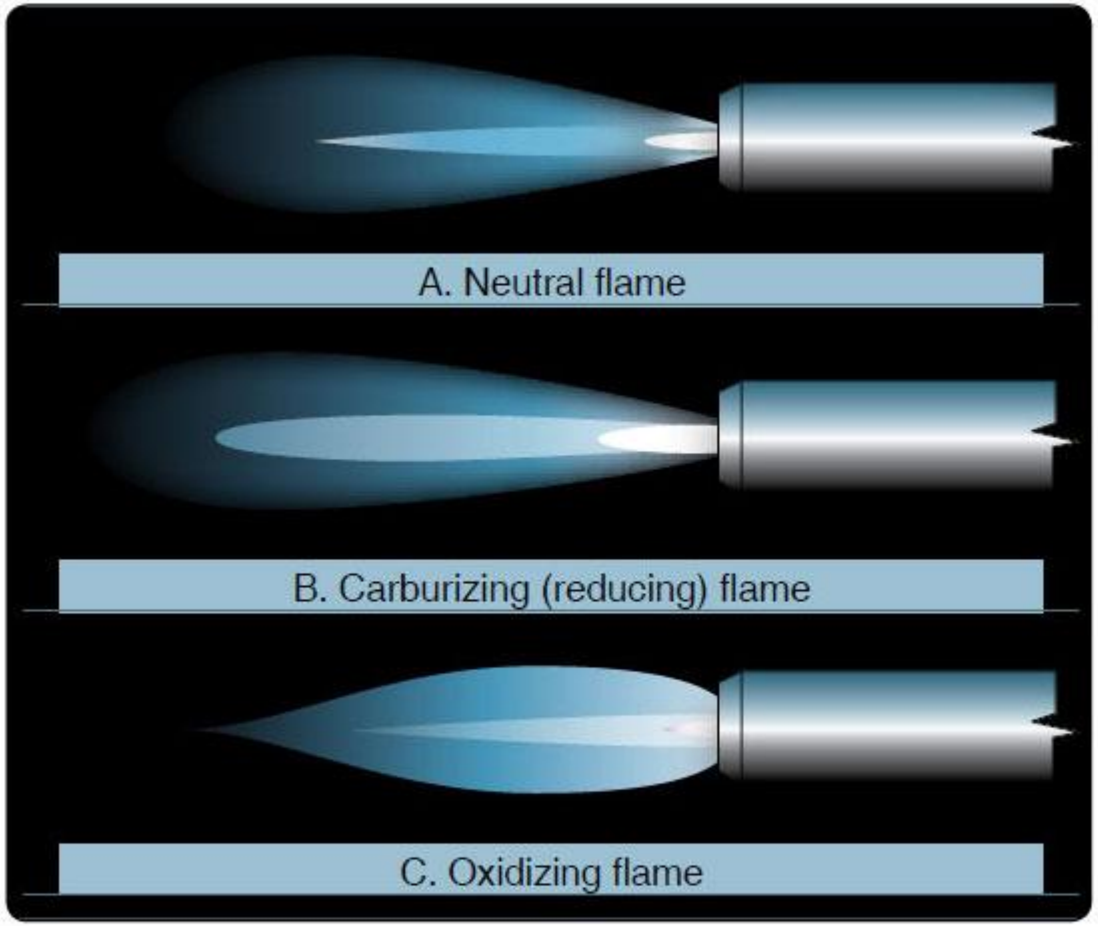


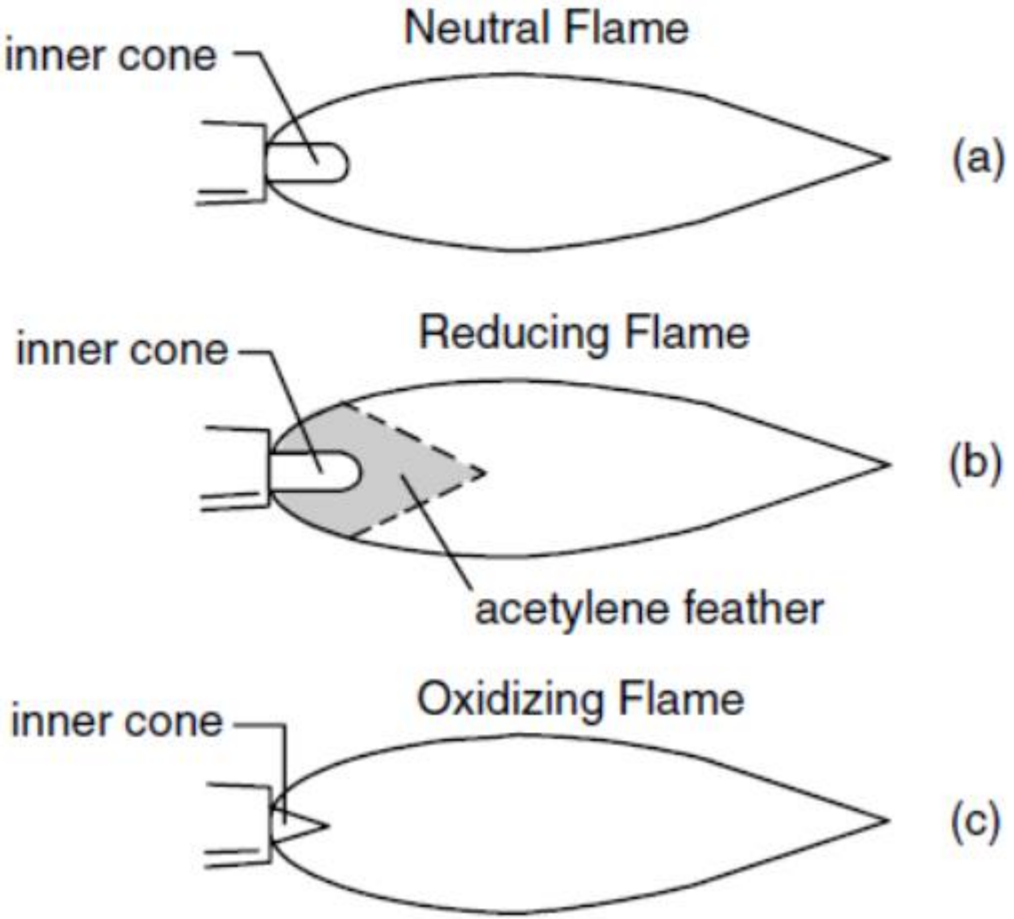
Gas Welding: Oxyacetylene welding

- Joins metals by heating them with a flame caused by the reaction between a fuel gas and oxygen
- A flux may be used to deoxidize and cleanse the weld metal
- The flux melts, solidifies, and forms a slag skin on the resultant weld metal
- Types of Flame in Oxy Acetylene Gas Welding
  - Neutral flame
  - Oxidizing flame
  - Carburizing (Reducing) flame

[Acetylene Flame Types - YouTube]

Types of Flames in Oxyacetylene welding





Types of Flames in Oxyacetylene welding

- Neutral flame:- case where oxygen (O_2) and acetylene (C_2H_2) are mixed in equal amounts and burned at the tip of the welding torch
  - A short inner cone and a longer outer envelope characterize a neutral flame. The inner cone is the area where the primary combustion takes place through the chemical reaction between O_2 and C_2H_2
  - Applications: Mostly Metals
- Oxidizing flame:- excess acetylene is used, the resulting flame is called a reducing flame
  - Combustion of acetylene is incomplete, As a result, a greenish acetylene feather between the inert cone and the outer envelope characterizes a reducing flame
  - Applications: Aluminum and its alloys
- Carburizing (Reducing) flame:- excess oxygen is used, the flame becomes oxidizing because of the presence of unconsumed oxygen. A short white inner cone characterizes an oxidizing flame
  - Applications: Brass parts.

Advantages and Disadvantages of Gas welding

Advantages

- Oxy-Fuel gas can be easily controlled.
- Suitable for thin sheets.
- Equipment is portable.
- It can weld most common materials.

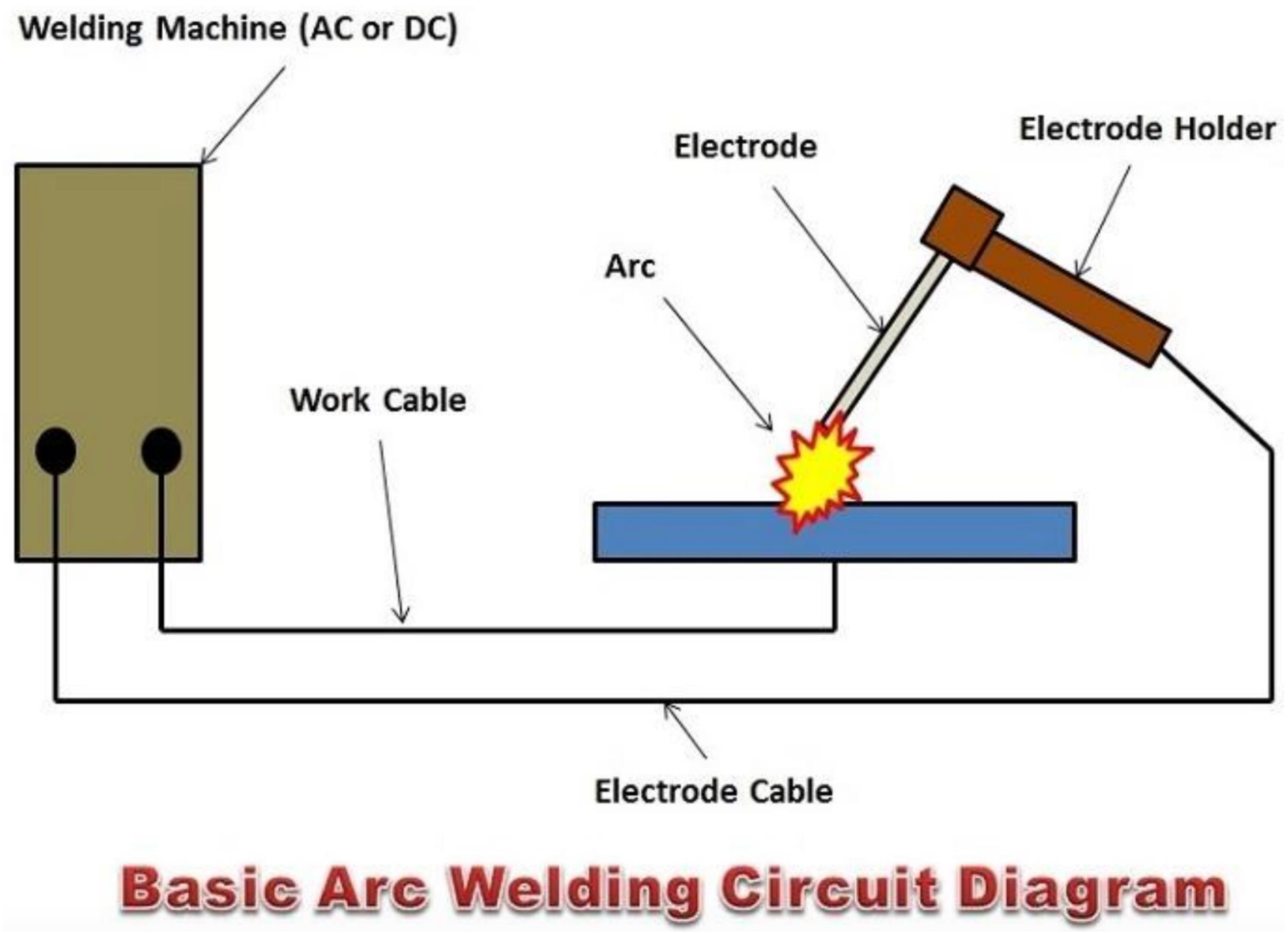
Disadvantages

- Heavy sections cannot be joined.
- Flame temperature is less than that of arc.
- Fluxes used produce fumes that are irritating to eyes, nose and lungs.
- Slower than arc welding Process.

Arc welding

- Method of fusion welding in which the metals at the joint is heated to molten state by an electric arc.
- Arc column is generated between an anode(electrode) and the cathode(metal to be joined)
- Temperature of the arc is about 6000ºC to 7000ºC

Arc Welding



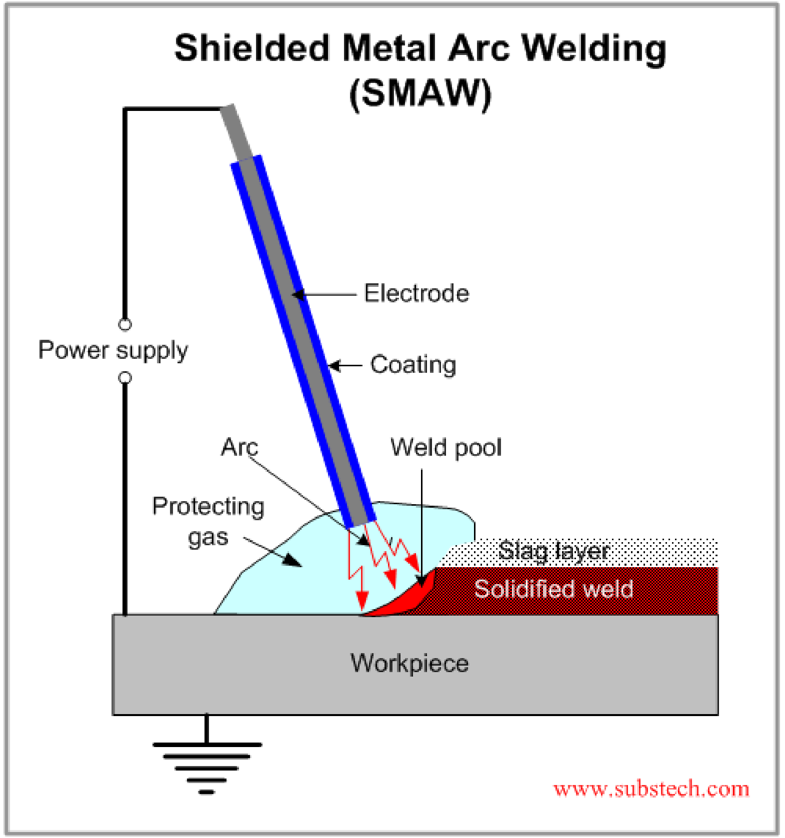
Welding Machine (AC or DC)
Electrode
Electrode Holder
Arc
Work Cable
Electrode Cable
Basic Arc Welding Circuit Diagram

Electrodes

- **Non- consumable electrodes**: May be made of carbon, tungsten or graphite which do not consume during welding operation.
- **Consumable electrodes**: It is consumed during welding operation. May be made of various metals depending upon the purpose and chemical composition of the metals to be welded.
- These electrode may be bare(uncoated) or flux coated

Arc Welding

- The heat is generated by an electric arc between base metal and a consumable electrode.
- In this process electrode movement is manually controlled hence it is termed as manual metal arc welding
- In shielded metal arc welding, the protection to the weld pool is provided by covering of
  - Slag formed over the surface of weld pool/metal and
  - inactive gases generated through thermal decomposition of flux/coating materials on the electrode

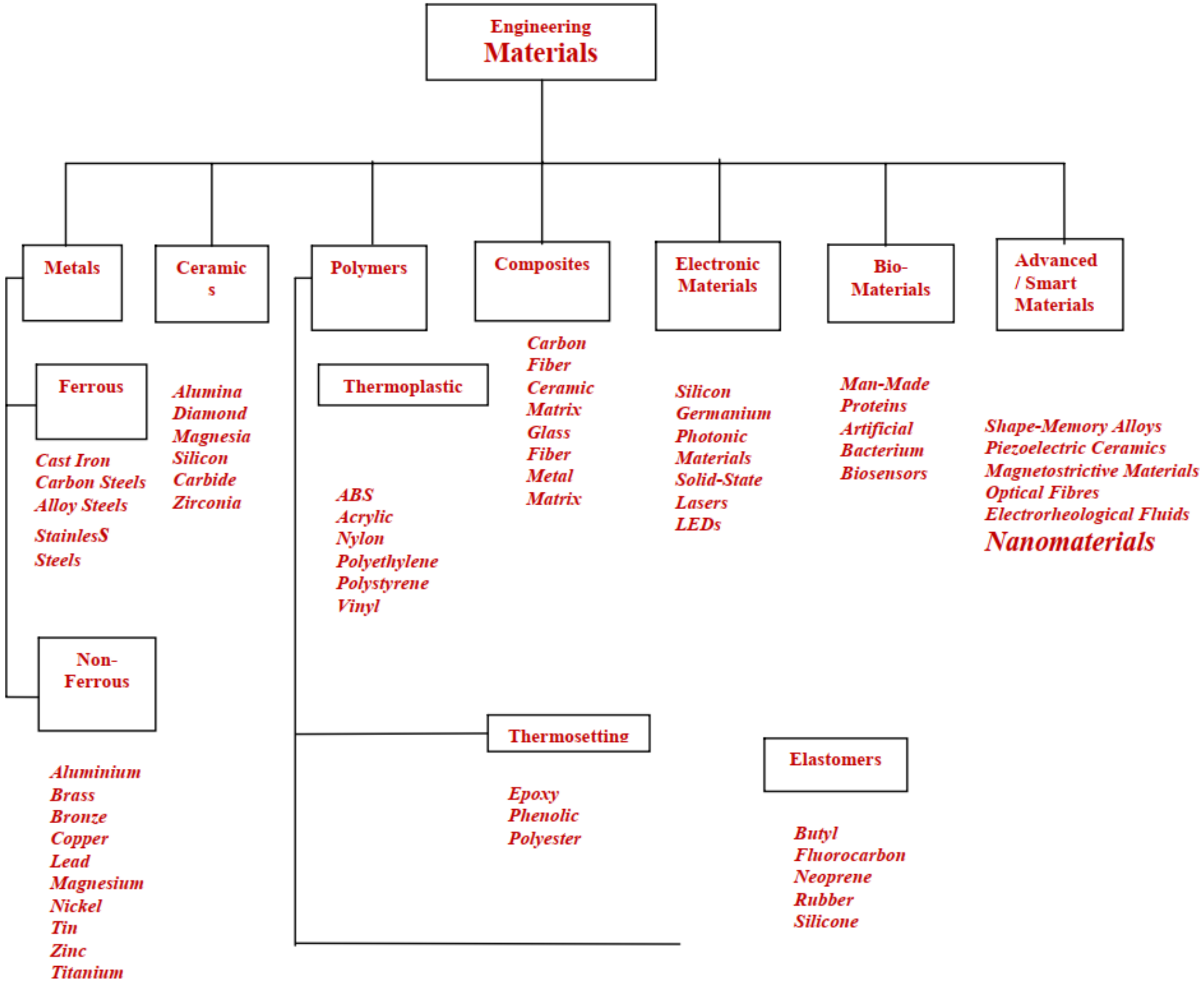


Engineering Materials

Engineering Materials

- **Engineering Material**: Part of inanimate matter, which is useful to engineer in the practice of his profession (used to produce products according to the needs and demand of society)
- **Material Science**: Primarily concerned with the search for basic knowledge about internal structure, properties and processing of materials and their complex interactions/relationships
- **Material Engineering**: Mainly concerned with the use of fundamental and applied knowledge of materials, so that they may be converted into products, as needed or desired by the society (bridges materials knowledge from basic sciences to engineering disciplines)

Engineering Materials



Composite Materials

- Any combination of two or more different materials at macroscopic level
- Two inherently different materials that when combined together produce a material with properties that exceed constituent materials
  - Eg: Reinforcement phase (eg. Fibers)
  - Binder phase (eg. Compliant matrix)
- Advantages:
  - High strength and stiffness
  - Low weight ratio

Smart Materials

- Materials that are manipulated to respond in a controllable and reversible way, modifying some of their properties as a result of external stimuli
- For example, we can talk about sportswear with ventilation valves that react to temperature and humidity by opening when the wearer breaks out in a sweat and closing when the body cools down
- Types
  - Piezoelectric materials
  - Shape memory materials
  - Magnetorheological materials

Additional Resources

- Smart materials: https://www.youtube.com/watch?v=aV07hCF7-AQ
- Composits: 'Composites technology to play vital role in space programmes' (business-standard.com)
- The Role of Composites in Aerospace Engineering (azom.com)
- Self-assembling material pops into 3D (youtube.com)

Thank You